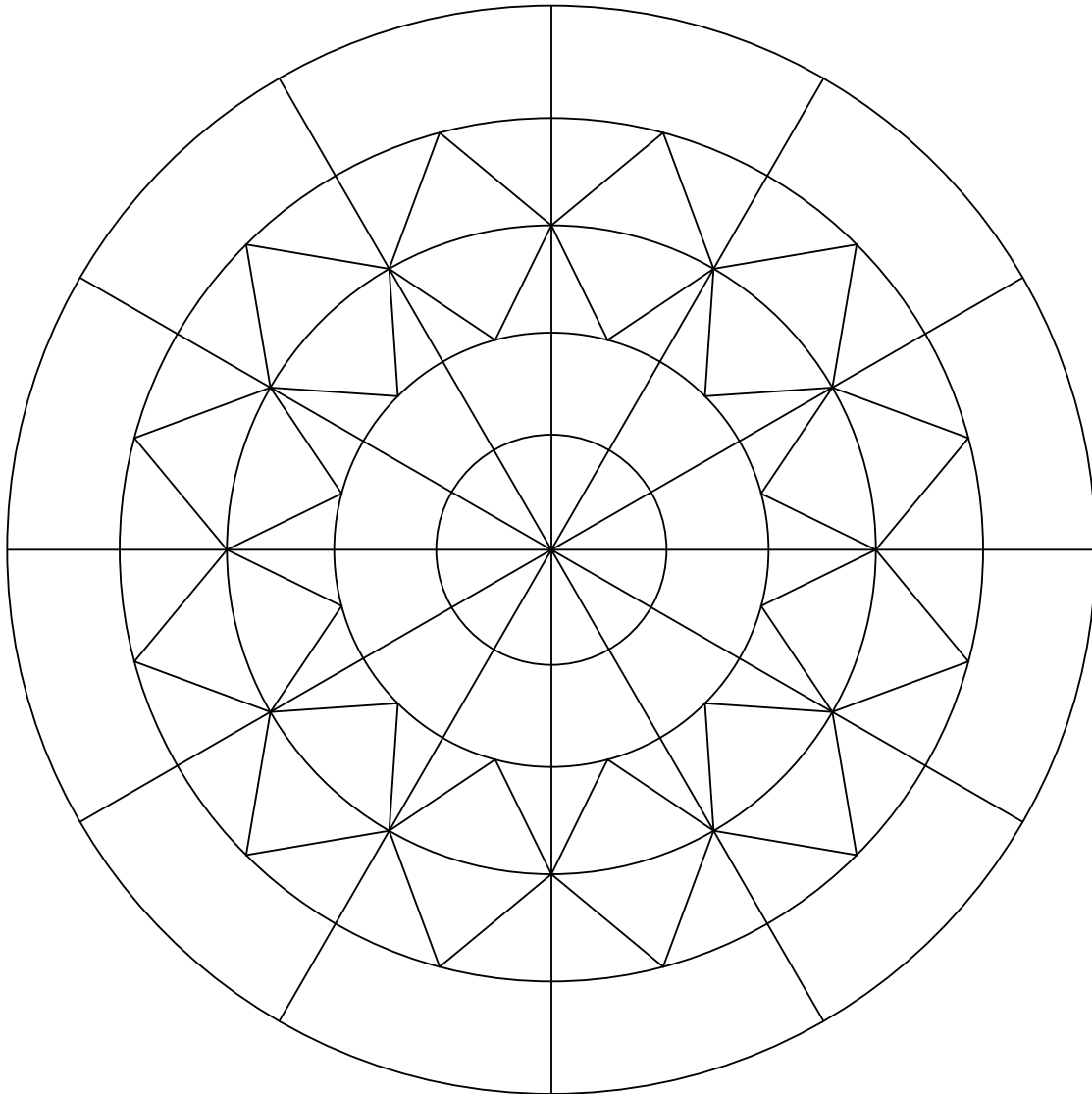


Color a turning rosette

Choose a color pattern for one slice. Repeat it all the way around.

Can you color the design so it looks the same after a quarter-turn?

Try another copy with a different turning symmetry.



My design matches after a turn of _____ degrees.

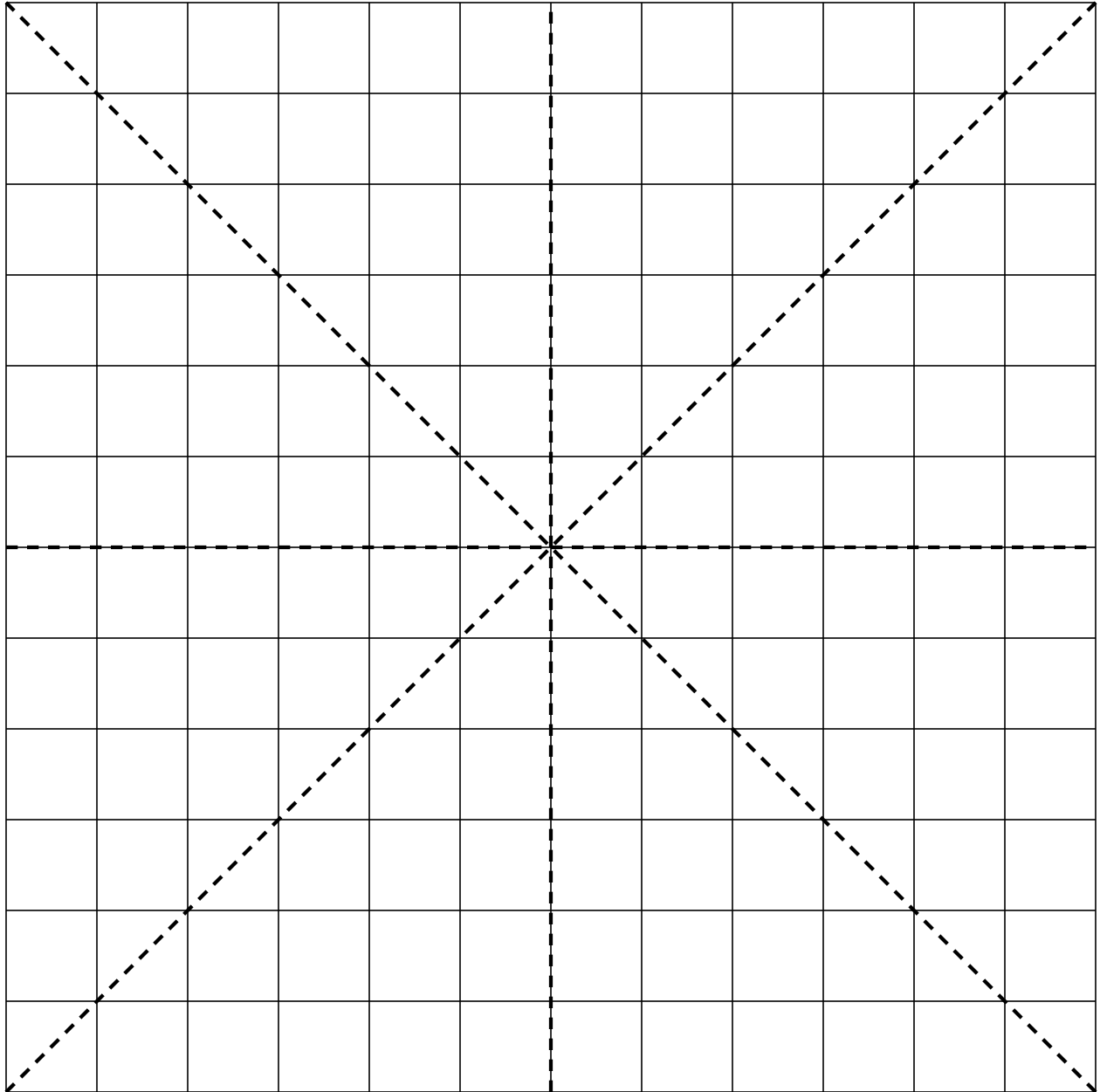
Can you find a mirror line? Draw it lightly with a ruler.

Make a square kaleidoscope

Color one triangular sector. Reflect its colors across each dashed mirror.

Keep going until the whole square is colored. A tracing sheet can help.

Challenge: make a second coloring that keeps quarter-turns but loses mirrors.



Every reflection should match colors as well as shapes.

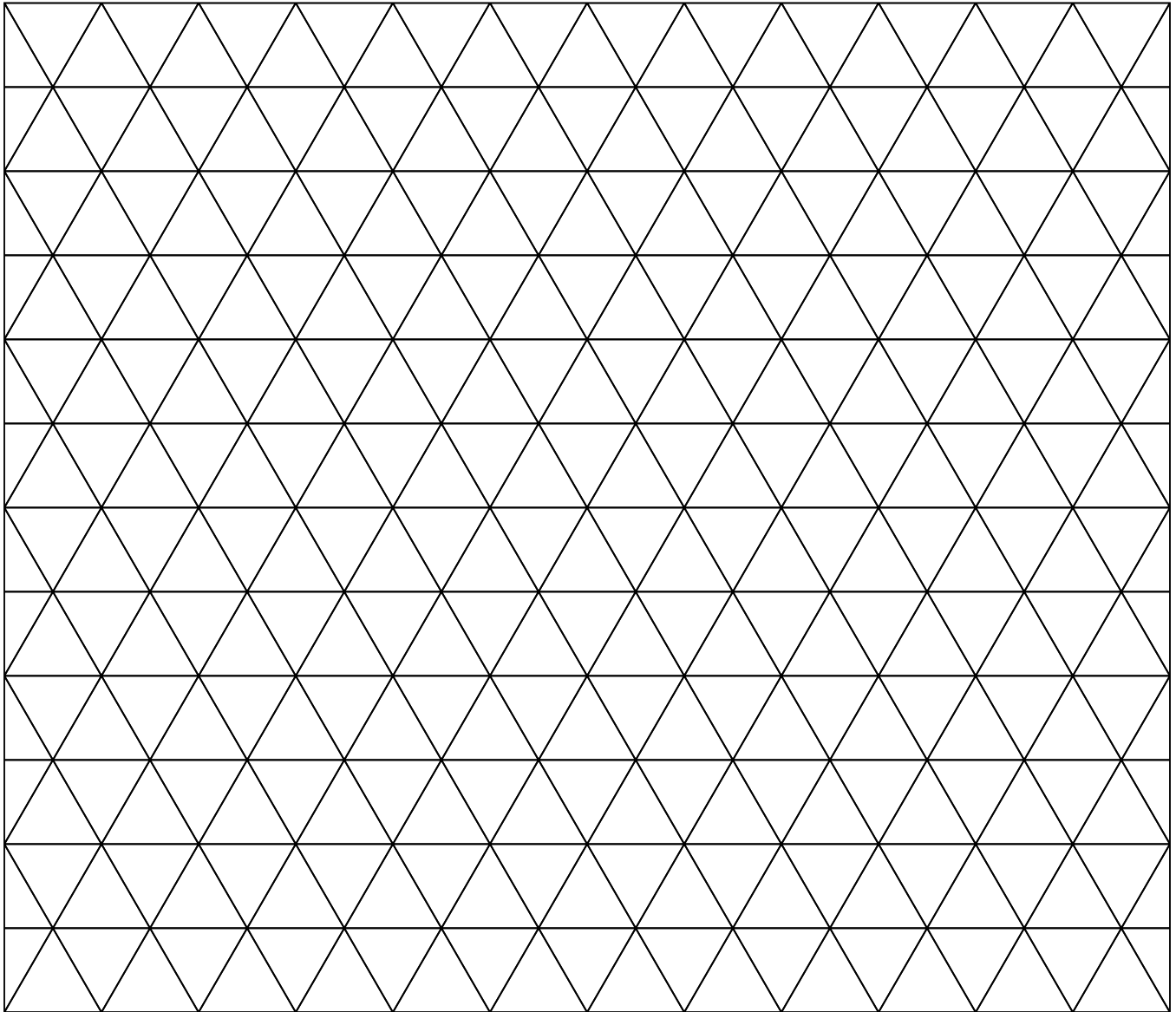
What is the smallest colored piece that tells you the whole design?

Triangles that tile the page

Color a repeating pattern. Try stripes, stars, or large triangles.

Challenge: use only two colors, with different colors across every shared edge.

Where can you slide your pattern so it matches itself?

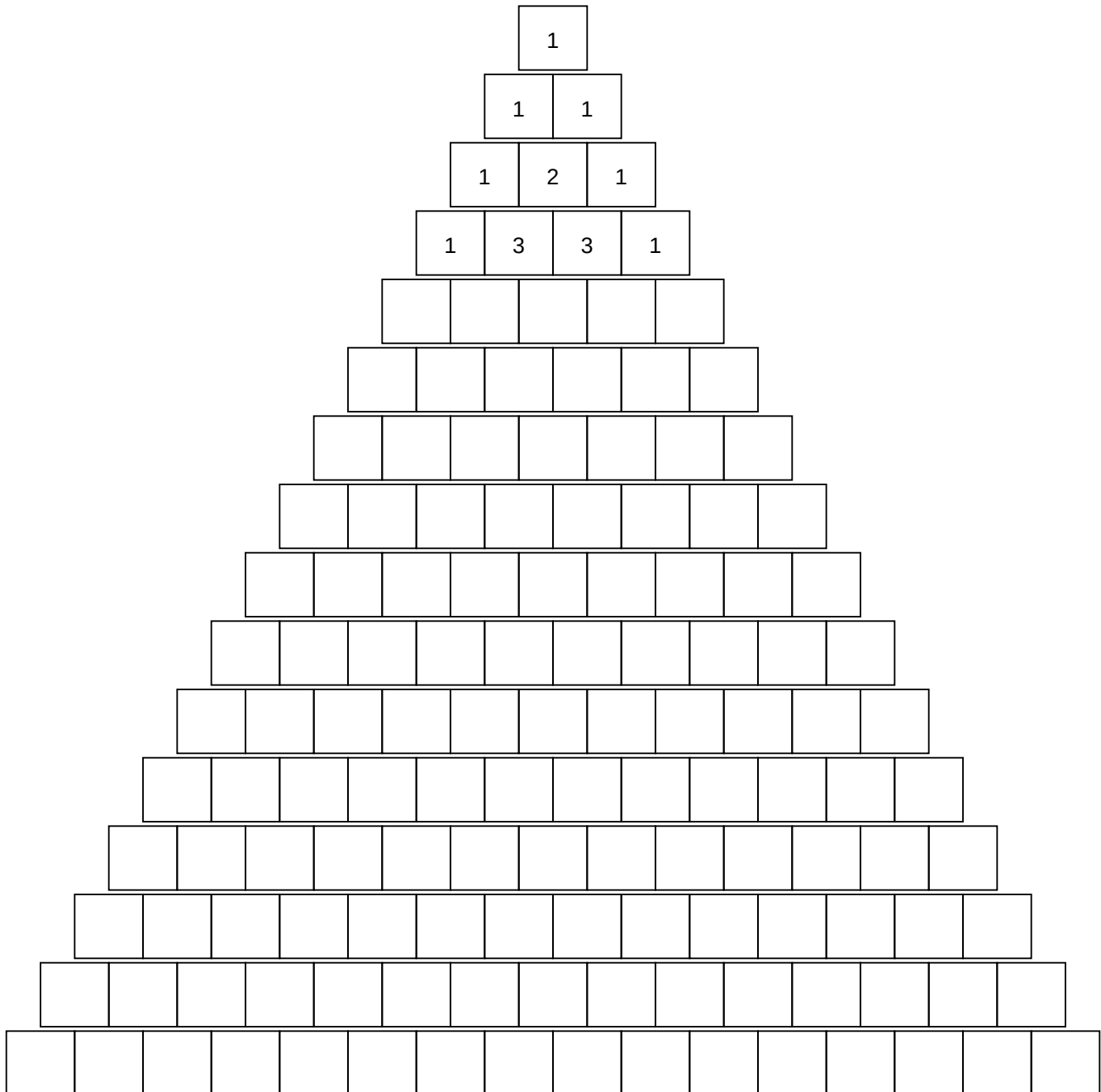


Find a mirror line in the uncolored tiling. Does your coloring keep it?

A finite page ends, but imagine your pattern continuing forever.

Pascal triangle: color the odd ones

Start each row and end each row with 1. Each other box is the sum of the two boxes just above it. Fill the empty boxes; then color odd numbers. Leave even numbers white. What shapes appear?

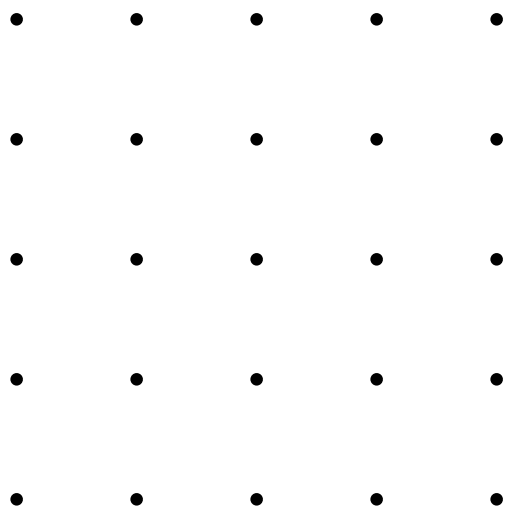


Shortcut: odd + odd is even; even + even is even; odd + even is odd.
Use O and E instead of large numbers if you prefer.

Dots and Boxes

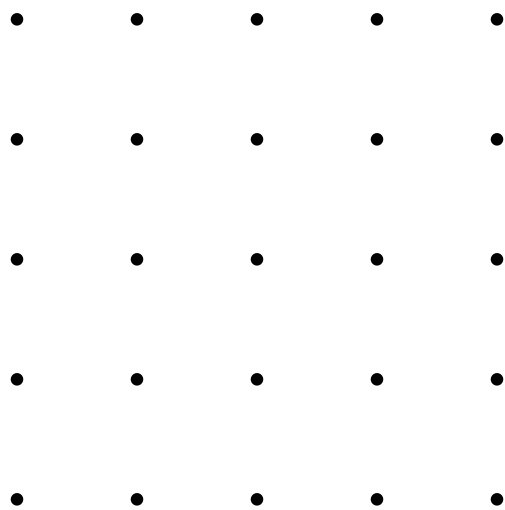
Take turns joining neighboring dots with one horizontal or vertical line.
Complete a small square? Write your initial inside it and take another turn.
When every box is claimed, the player with the most boxes wins. Ties are possible.

GAME 1



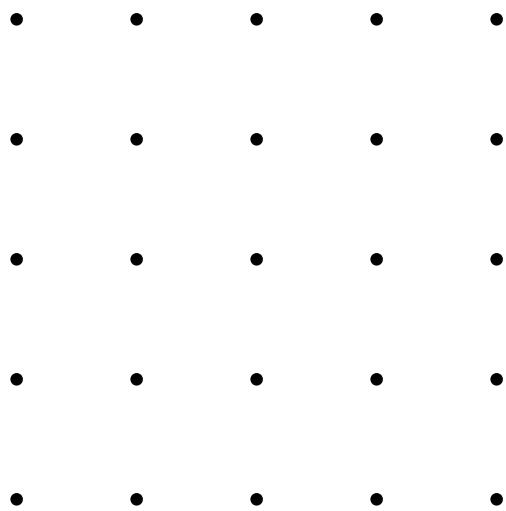
Scores: _____ / _____

GAME 2



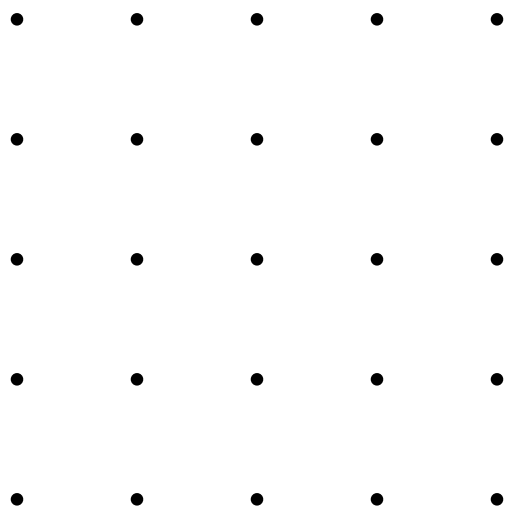
Scores: _____ / _____

GAME 3



Scores: _____ / _____

GAME 4



Scores: _____ / _____

Sprouts: draw the last curve

Start with the dots in a play area. Take turns drawing one curve between two dots, or from a dot back to itself. Add one NEW dot along your new curve. Curves cannot cross or touch other curves or dots except at their endpoints. At most three curve ends may meet any dot. A loop uses two ends at its dot. If you cannot make a legal move, you lose. Try two dots before three.

GAME 1 / 2 starting dots



GAME 2 / 3 starting dots



Explore: Does the first player always win? Try changing your opening move.

Conway's Game of Life

Shade living cells; leave dead cells white. Each cell has EIGHT neighbors: above, below, left, right, and the four diagonals. Do not count the cell itself.

A living cell survives with 2 or 3 living neighbors; otherwise it dies.

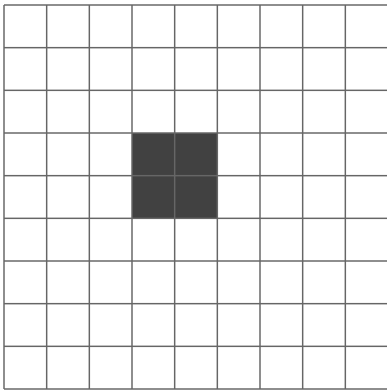
A dead cell becomes living with exactly 3 living neighbors.

All changes happen together. Count on the OLD grid; draw on the NEW grid.

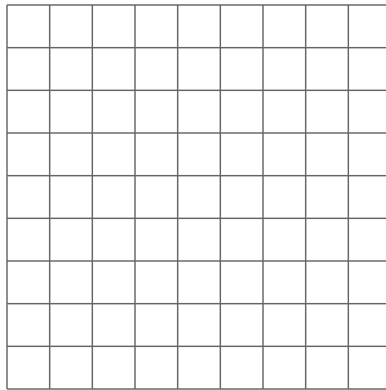
The world continues beyond each grid. Start with all unshown cells dead.

A. Block

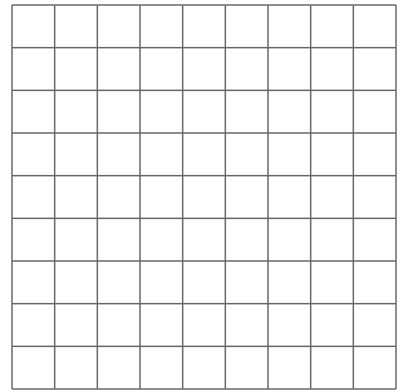
Start



Generation 1

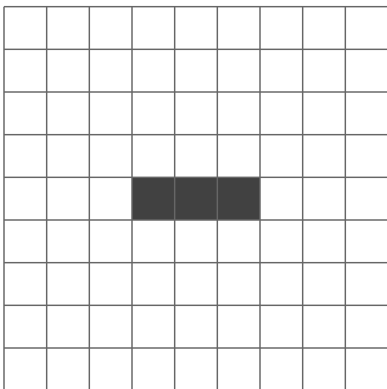


Generation 2

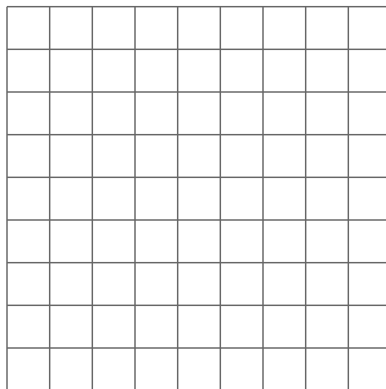


B. Blinker

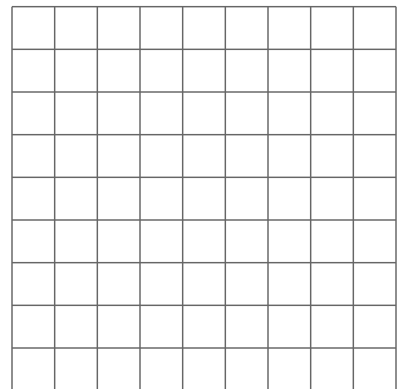
Start



Generation 1



Generation 2

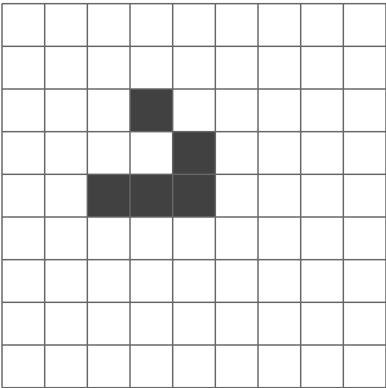


Which pattern stays still? Which changes and then comes back?

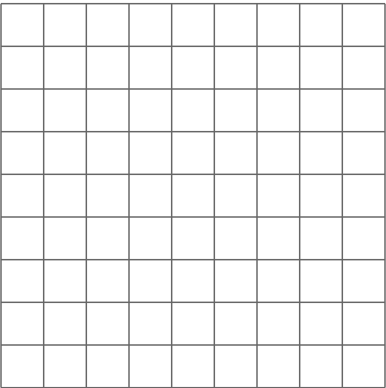
Follow a glider

Use the rules on page 7. Copy each new generation into the next grid.
Count all eight neighbors using only the previous generation.
After four steps, compare the shape AND its position with the start.

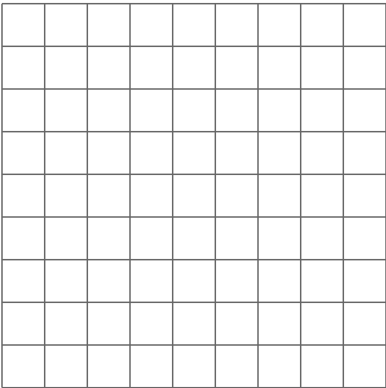
Start



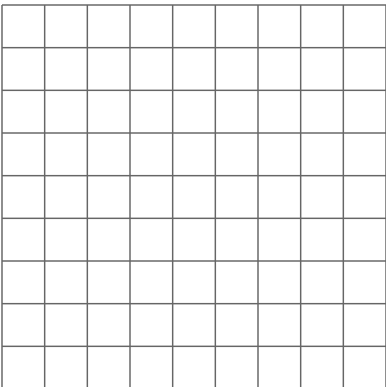
Generation 1



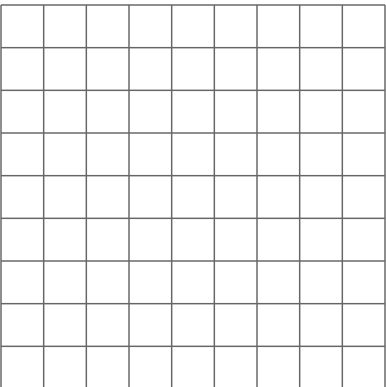
Generation 2



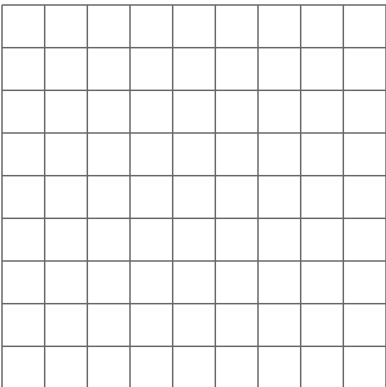
Generation 3



Generation 4



Extra: generation 5



What changed? _____

Prediction: Where will the glider be after eight steps?

Invent a Life experiment

Start with 3 to 6 living cells near the center of the first grid.
Predict what will happen. Then compute three generations to test your guess.
If the pattern reaches an edge, use more graph paper; do not wrap it around.

My start

Generation 1

Generation 2

Generation 3

My prediction was _____

What I noticed _____

Leader notes and answer key

Coloring: many solutions are possible. Ask children to explain a rule they chose.

p. 1: repeating every third slice preserves a quarter-turn (90 degrees).

p. 2: mirror matching gives fourfold turning symmetry. For the challenge, place the same small asymmetric motif in four quarter-turn positions.

p. 3: color upward triangles one color and downward triangles another.

p. 4: odd entries form a triangular fractal pattern. The first rows of parity are:

```

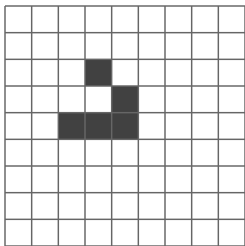
      0
    0 0
  0 E 0
0 0 0 0
  0 E E E 0
0 0 E E 0 0
  0 E 0 E 0 E 0
0 0 0 0 0 0 0 0
  
```

p. 7: the block stays unchanged. The blinker alternates vertical and horizontal.

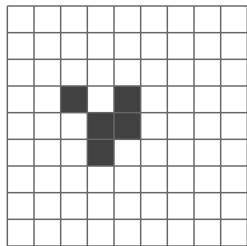
p. 8: the glider returns to its shape one cell down and one cell right after 4 steps.

After 8 steps it has moved two cells down and two cells right.

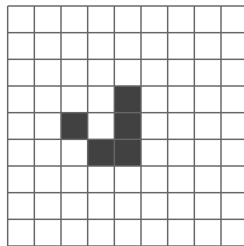
Glider 0



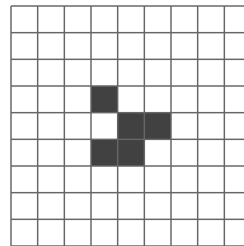
Glider 1



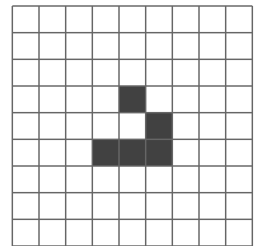
Glider 2



Glider 3



Glider 4



p. 9: outcomes depend on the chosen seed. Check cells that start dead too!

Common Life mistake: updating one grid in place changes the rules.

For Sprouts, a new dot has two ends already, so it has room for only one more.

These activities introduce examples; they are not classifications or proofs.

Rules and further reading:

Conway Life: conwaylife.com/wiki/Rulestring

Sprouts: nrich.maths.org/games/sprouts

Dots and Boxes: mathforlove.com (Games to Play at Home, p. 2)

All worksheet text, layouts, and drawings in this pack were newly prepared.

You may print, adapt, and share this original pack for Science Friday.